Horizontal launches

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Find flight time from vertical motion and then horizontal range.

SUCCESS CRITERIA - I can...

1. Separate vertical and horizontal givens.
2. Find or use fall time before calculating range.
3. Check units and interpret the landing point.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “horizontal” and “vertical” to explain your model or result. Sentence starters are optional.

horizontal: along a level line, usually the x direction

vertical: up and down, usually the y direction

displacement: change from initial position to final position, with direction

01 Warm-up: bring back an earlier idea

A spherical planet has surface field 75 N/kg.
Find the field at center distance 5 R, where R is the planet radius.

Word help: radius = distance from the center to the edge of a circle or sphere

Sentence starter: My design shows … . It meets the requirement by … .

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $h = \frac{1}{2}gt^2$; range = $v_x t$.
Horizontal launch; h is the positive drop, t is fall time; g = 10 m/s². CPA may use the supplied time.

Worked reasoning: A horizontal launch falls 20 m with g=10 m/s². From $20=5\,t^2$, t=2 s. At horizontal speed 3 m/s, range=6 m.

Your turn: Draw a diagram labeling height and range. Number the two solution steps to show why time comes first.

Sentence starter: My claim is … . The diagram/data show …, so … .



Add velocity / acceleration arrows.

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

On the diagram, label height h, horizontal range and initial horizontal velocity. Which direction determines the fall time?

Sentence starter: I found ... [unit]. This value means ... in this situation.



Add velocity / acceleration arrows.

Marzano check - circle: ① ② ③

04 Apply the physics

In a launch simulation, two balls leave the same platform horizontally. Both fall for 2 s. One launches at 4 m/s and the other at 6 m/s.
Draw and label both paths; calculate each horizontal range.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Add velocity / acceleration arrows.

Marzano check - circle: ① ② ③

05 Explain your reasoning

Two platforms have heights 5 and 20 m. Supplied fall times are 1 and 2 s. At the same launch speed, does quadrupling height quadruple range?
Explain.

Sentence starter: My claim is ... . The diagram/data show ..., so ... .

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A 6 kg cart starts from rest under 48 N net force right for 5 s.
Find acceleration and final velocity.
State one assumption.
Label the given and predicted quantities on the model.

Word help: net force = the vector sum of all forces on one object

Sentence starter: I predict ... . My model supports this because ... .

Velocity (m/s)



Time (s)

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

A horizontal launch at 5 m/s stays in air 3 s. Where does it land relative to the platform base?
Label the model with the known quantities before calculating. Ignore air resistance; use g = 10 m/s² when needed.

Sentence starter: The … in my model represents … . Its label shows … .

Separate horizontal and vertical motion

Label x and y; state positive directions.

Velocity:

Acceleration / force:

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A target is 10 m from a platform whose fall time is 2 s. **Choose** the horizontal launch speed, then predict where half that speed lands.
Label the model with the known quantities before calculating. Ignore air resistance; use g = 10 m/s² when needed.

Sentence starter: I predict … . My model supports this because … .



Add velocity / acceleration arrows.

Marzano check - circle: 1 2 3